

Lecture 33. Aggregation

Aggregation

Aggregate Functions



So far, all SQL expressions have referred to the values in a single row at a time

```
[expression] as [name], [expression] as [name], ...
```

```
select [columns] from [table] where [expression] order by [expression];
```

An aggregate function in the [columns] clause computes a value from a group of rows

```
create table animals as
select "dog" as kind, 4 as legs, 20 as weight union
select "cat"         , 4         , 10         union
select "ferret"      , 4         , 10         union
select "parrot"       , 2         , 6          union
select "penguin"     , 2         , 10         union
select "t-rex"       , 2         , 12000;
```

```
select max(legs) from animals;
```

max(legs)

4

animals:

kind	legs	weight
dog	4	20
cat	4	10
ferret	4	10
parrot	2	6
penguin	2	10
t-rex	2	12000

```
select max(legs) from animals;
select sum(weight) from animals;
select max(legs-weight) + 5 from animals;
.....
```

other aggregate functions:

- count
 - count(distinct animals)

- avg

Selecting Groups



Rows in a table can be grouped, and aggregation is performed on each group

```
[expression] as [name], [expression] as [name], ...
```

```
select [columns] from [table] group by [expression] having [expression];
```

A **having** clause filters the set of groups that are aggregated

```
select weight/legs, count(*) from animals group by weight/legs having count(*)>1;
```

animals:

kind	legs	weight
dog	4	20
cat	4	10
ferret	4	10
parrot	2	6
penguin	2	10
t-rex	2	12000

weight/legs	count(*)
5	2
2	2

weight/legs=5

weight/legs=2

weight/legs=2

weight/legs=3

weight/legs=5

weight/legs=6000